

Handling Multiple Baselines in ADaM: A Practical Approach

Consistent Baseline. Clear Rules. Reliable Results.

Overview

- In complex clinical trial designs, such as crossover studies, re-baselining after washout and treatment re-initiations subjects frequently require more than one baseline. Improper baseline handling can lead to incorrect baseline-dependent results, inconsistent interpretation, increased validation effort, and potential regulatory queries.
- This guidance addresses common programming workarounds (such as inventing non-standard baseline parameter codes or missing flags) by demonstrating a consistent approach using BASETYPE together with standard ABLFL and BASE conventions.
- Inconsistent baseline handling can lead to biased results and regulatory findings.
- ADaM requires a clear, rule-based approach to define and flag the correct baseline for each analysis.
- The approach assigns the applicable BASETYPE, identifies eligible pre-dose candidates and selects one record with ABLFL = "Y" for each subject, parameter, and BASETYPE when an eligible baseline exists.

Why It Matters

- Ensures correct change from baseline.
- Reduces risk of regulatory queries.
- Supports transparent & reproducible analysis.

Common Multiple Baseline Scenarios

1. Crossover Study

A new baseline is established at the start of each treatment period. Subjects receive **Treatment A** during Period 1, followed by a washout period, and then receive **Treatment B** during Period 2. To accurately assess treatment effects within each period, a separate baseline is defined before the initiation of each treatment period.

Sequence: Period 1 (Treatment A) → Washout → Period 2 (Treatment B).

BASETYPE = 'PERIOD 1' at start of Period 1; BASETYPE = 'PERIOD 2' at start of Period 2.

Example:

USUBJID	PARAMCD	APERIOD	TRTA	AVISIT	AVAL	BASETYPE	ABLFL	BASE	CHG
101	SYSBP	1	Treatment A	Screening	122	PERIOD 1		120	
101	SYSBP	1	Treatment A	Period 1 Day 1 Pre-dose	120	PERIOD 1	Y	120	0
101	SYSBP	1	Treatment A	Period 1 Day 1 Post-dose	114	PERIOD 1		120	-6
101	SYSBP	1	Treatment A	Period 1 Day 7	116	PERIOD 1		120	-4
101	SYSBP	2	Treatment B	Period 2 Day 1 Pre-dose	125	PERIOD 2	Y	125	0
101	SYSBP	2	Treatment B	Period 2 Day 1 Post-dose	117	PERIOD 2		125	-8
101	SYSBP	2	Treatment B	Period 2 Day 7	119	PERIOD 2		125	-6

2. Two-Phase Parallel Study

A new baseline is established during the Extension Phase.

Sequence: In the Initial Phase, subjects receive either Treatment A or Placebo. Upon entering the Extension Phase, all subjects receive Treatment A, with subjects previously assigned to Treatment A continuing their treatment and subjects previously assigned to Placebo switching to Treatment A.

BASETYPE = 'INITIAL BASELINE' in the Initial Phase and BASETYPE = 'EXTENSION BASELINE' in the Extension Phase.

Example:

USUBJID	PARAMCD	PHASE	TRTA	AVISIT	AVAL	BASETYPE	ABLFL	BASE	CHG
101	SYSBP	INITIAL	DRUG A	Screening	110	INITIAL BASELINE	Y	110	0
101	SYSBP	INITIAL	DRUG A	VISIT 1	120	INITIAL BASELINE		110	10
101	SYSBP	INITIAL	DRUG A	VISIT 2	115	INITIAL BASELINE		110	5
101	SYSBP	INITIAL	DRUG A	VISIT 3	122	INITIAL BASELINE		110	12
101	SYSBP	EXTENSION	DRUG A	VISIT 4	125	EXTENSION BASELINE	Y	125	0
101	SYSBP	EXTENSION	DRUG A	VISIT 5	124	EXTENSION BASELINE		125	-1
101	SYSBP	EXTENSION	DRUG A	VISIT 6	116	EXTENSION BASELINE		125	-9
102	SYSBP	INITIAL	Placebo	Screening	125	INITIAL BASELINE	Y	125	0
102	SYSBP	INITIAL	Placebo	VISIT 1	116	INITIAL BASELINE		116	0
102	SYSBP	INITIAL	Placebo	VISIT 2	115	INITIAL BASELINE		116	-1
102	SYSBP	INITIAL	Placebo	VISIT 3	118	INITIAL BASELINE		116	2
102	SYSBP	EXTENSION	DRUG A	VISIT 4	120	EXTENSION BASELINE	Y	120	0
102	SYSBP	EXTENSION	DRUG A	VISIT 5	121	EXTENSION BASELINE		120	1
102	SYSBP	EXTENSION	DRUG A	VISIT 6	118	EXTENSION BASELINE		120	-2

3. Treatment Re-initiation

A new baseline is established at the time of treatment re-initiation. Subjects initially receive **Treatment A 100 mg** but discontinue treatment due to a predefined reason. Upon restarting treatment, subjects re-initiate **Treatment A at 50 mg**, and a new baseline is defined to support the evaluation of efficacy and safety outcomes following treatment re-initiation.

Sequence: Treatment → Off Treatment → Treatment.

BASETYPE = 'INITIAL BASELINE' at first treatment; BASETYPE = 'RE-INITIATION BASELINE' at restart.

Example:

USUBJID	PARAMCD	AVAL	PHASE	BASETYPE	ABLFL	BASE	CHG
101	SYSBP	120	Initial	INITIAL BASELINE	Y	120	0
101	SYSBP	119	Initial	INITIAL BASELINE		120	-1
101	SYSBP	123	Initial	INITIAL BASELINE		120	3
101	SYSBP	125	Re-initiation	RE-INITIATION BASELINE	Y	125	0
101	SYSBP	123	Re-initiation	RE-INITIATION BASELINE		125	-2
101	SYSBP	118	Re-initiation	RE-INITIATION BASELINE		125	-7

Practical Approach for Multiple Baseline Handling

Step 1: Assign Baseline Type (BASETYPE)

Assign the appropriate **BASETYPE** to every record based on the treatment segment, period, or re-baselining scenario to which the record belongs.

Step 2: Identify Eligible Baseline Records

Identify all pre-dose assessment records within each treatment segment or analysis period. These records are potential baseline candidates.

Step 3: Select One Baseline per BASETYPE

Within each subject, parameter, and BASETYPE combination, identify a single baseline record and set **ABLFL** = "Y". This record represents the reference point for analysis within that baseline type.

Step 4: Derive Analysis Variables

Calculate **BASE**, **CHG**, and **PCHG** using only the selected baseline record within the corresponding BASETYPE.

- **BASE** = Baseline value
- **CHG** = Analysis Value – BASE
- **PCHG** = ((Analysis Value – BASE) / BASE) × 100

Key Variables

BASETYPE: Identifies the baseline against which a record is evaluated.

ABLFL: Flags the selected baseline record within each BASETYPE.

BASE: Baseline value associated with the selected baseline record.

CHG: Change from baseline.

PCHG: Percent change from baseline.

Pitfalls to Avoid

1. Creating Non-Standard Baseline Parameters

Do not create new parameter codes (for example, *BASE2* or *BASE3*) to represent additional baselines. Multiple baselines should be handled using BASETYPE, while retaining the original parameter definitions.

2. Creating Additional Baseline Columns

Avoid deriving separate variables such as *BASE2* or *CHG2*. Instead, represent each baseline scenario as a separate BASETYPE and derive BASE, CHG, and PCHG within that context.

3. Assigning Multiple Baseline Flags Within a BASETYPE

Within a subject, parameter, and BASETYPE combination, only one record should be flagged with ABLFL = "Y". Multiple baseline flags can lead to ambiguity and inconsistent analysis results.

4. Using the Wrong Baseline for Change Calculations

Ensure that CHG and PCHG are calculated using the baseline corresponding to the applicable BASETYPE. Later treatment periods should not use a baseline from an earlier period.

5. Leaving BASETYPE Missing on Analysis Records

Assign BASETYPE consistently to all records within the baseline segment, not only to the baseline record itself. Missing BASETYPE values can make it difficult to identify the correct baseline and trace derivations.

Validation Checks

1. Verify Baseline Flag Assignment

Ensure that **exactly one record** is flagged with **ABLFL = "Y"** for each unique combination of **Subject, Parameter, and BASETYPE**.

2. Verify BASETYPE Population

Confirm that **BASETYPE is populated for all analyzed records** within the applicable treatment segment, period, phase, or re-baselining scenario.

3. Verify Baseline Consistency

Ensure that the **BASE value remains constant** for all records belonging to the same Subject, Parameter, and BASETYPE combination.

4. Verify Baseline Timing

Confirm that the selected baseline assessment occurs within the protocol-defined baseline window (for example, **Day -7 to Day -1**) and prior to the corresponding treatment start date.

5. Verify Change Derivations

Ensure that **CHG** and **PCHG** are derived using the **BASE value from the same BASETYPE** and are not linked to a baseline from a different treatment period or segment.

6. Verify Traceability

Confirm that each baseline record can be traced back to an eligible pre-dose assessment and that the derivation of **ABLFL, BASE, CHG, and PCHG** is fully documented.

7. Verify Eligible Baseline Selection

Ensure that the record flagged as baseline represents the correct eligible pre-dose assessment according to the study-specific baseline selection rule (for example, last non-missing value before treatment start).

8. Verify Missing Baseline Handling

Confirm that records without a valid baseline have **BASE**, **CHG**, and **PCHG** derived according to study specifications and that no incorrect baseline is assigned.

Conclusion

By combining standard ADaM variables such as BASETYPE, ABLFL, BASE, and CHG, we can correctly handle crossover studies, washout periods, treatment re-initiation scenarios.

This approach improves consistency, avoids incorrect change calculations, and provides clear traceability for reviewers.